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INSPIRED BY SUSTAINABILITY, BUILT FOR BEEKEEPING

A unisex utility boot designed for beekeepers, combining PF Flyers heritage with sustainable material innovation.



*Vizcom Rendering

INTRODUCING THE

PF FLYERS CLOVER

HERITAGE FTW

THE GROUNDER



Heritage footwear rooted in military relief and everyday utility, serving as the baseline shoe inspiration for this project

BRAND VALUES

THE PERSPECTIVE



A women-owned heritage footwear brand challenging convention and creating products for evolving user needs

MODERN APPLICATION

THE TRANSLATION



Extending that vision into **Beekeeping**, an underserved market where sustainability, protection, and utility meet

UNDERSTANDING THE BEEKEEPER

MORE THAN A HOBBY

Beekeepers are stewards of our environment

- Manage Pollination & Protect Honeybee Habitats
- Support Global Food Supply
- Promote Biodiversity
- Ensure Healthy Ecosystems

DESIGN OPPORTUNITY

A boot design addressing Beekeepers' specific needs that aligns with their **sustainability values** and can seamlessly transition into an everyday **lifestyle boot**

USER NEEDS

- Protection
- Secure Fit
- Comfort
- Breathability
- Sustainability
- Versatility





Bee sting reaction following bees entering shoe

Video

COMPETITOR ANALYSIS

BEEKEEPING BOOTS

WELLINGTON BOOT



STRENGTHS

- Tall (protective)
- Impenetrable
- Waterproof

LIMITATIONS

- Loose Upper Opening
- Limited Sustainability
- Bulky Silhouette

WORK BOOTS

ARIAT WOMEN'S WORK BOOT



STRENGTHS

- Durable
- Strong Traction
- All-Day Comfort

LIMITATIONS

- Industrial Work Focus
- Open-Collar Design
- Limited Sustainability

LIFESTYLE BOOTS

BLUNDSTONE CHELSEA BOOT



STRENGTHS

- Slip-On
- All-Day Comfort
- Versatile Use

LIMITATIONS

- Unsealed Opening
- Limited Sustainability
- General-Use Intent

DESIGN DIRECTION



How might we design a cotton-based beekeeping boot that balances protection, breathability, and versatility for everyday utility?

MOOD BOARD























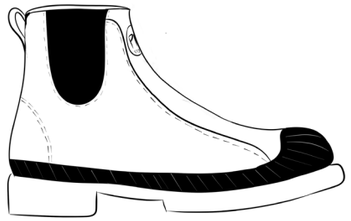


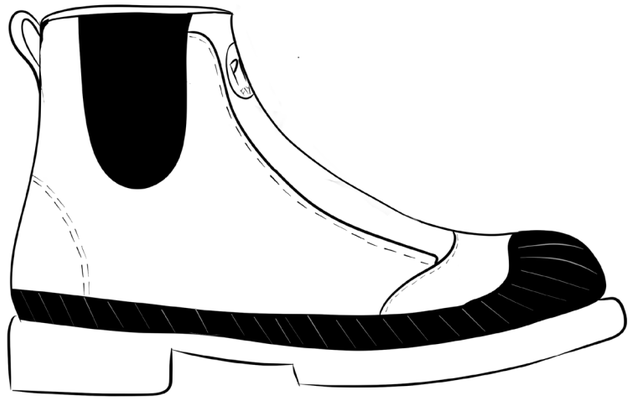
SKETCH IDEATION

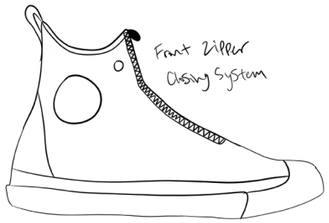






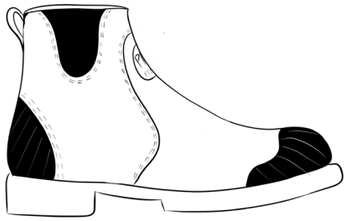




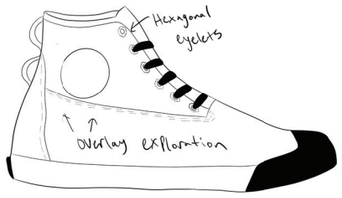


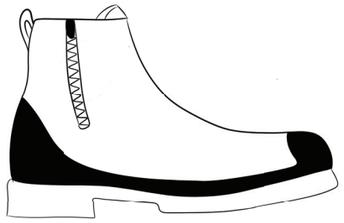


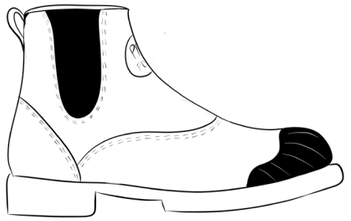


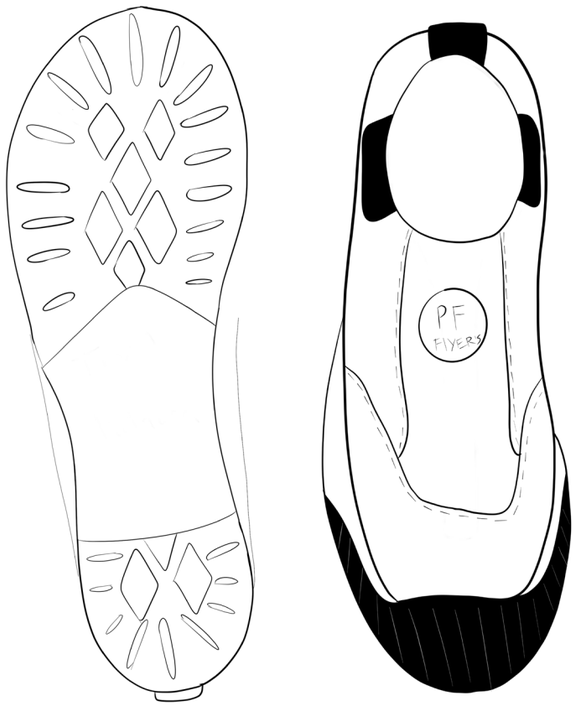






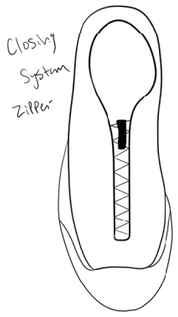


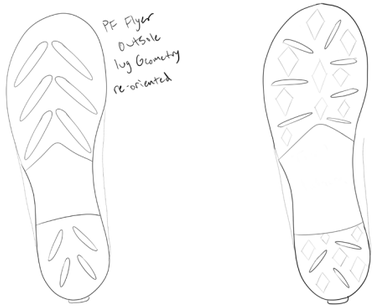














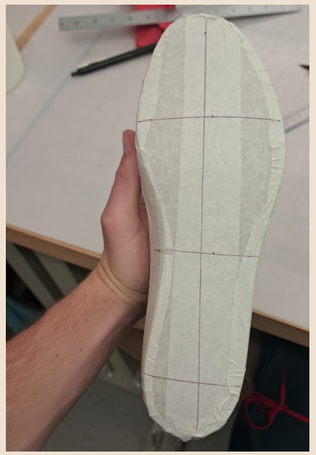


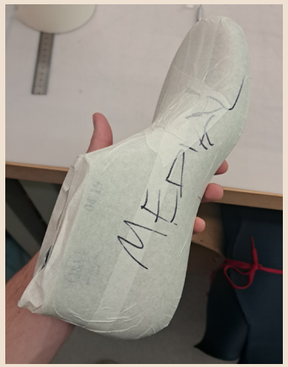
PROTOTYPING & ITERATION

PATTERN DEVELOPMENT

LAST TAPING









UPPER REFINEMENT

REFINING FORM & DETAILS













CAD MODELING

MIDSOLE & OUTSOLE



PF FLYERS
CLOVER

FEATURES & BENEFITS

- **DWR-COATED UPPER** for easy cleaning & enhanced durability
- **TPU TOE CAP & MUDGUARD** for abrasion & element protection
- **ADJUSTABLE STRAP CINCH** for a secure fit & sealing entry points
- **TREADED OUTSOLE** for reliable traction across varied outdoor surfaces
- **CHELSEA BOOT CONSTRUCTION** to enable quick slip-on access
- **PULL TABS** located on the heel to assist with easy donning & doffing



*Vizcom Rendering

MATERIALS

PULL TABS
KNIT - 100% COTTON

CINCH STRAP
KNIT - 100% COTTON

GUSSET
KNIT - ELASTIC GORE

PF PATCH
LEATHER

UPPER
WOVEN - 100% ORGANIC COTTON CANVAS
DWR ALUMINUM OXIDE + LANOLIN COATING

LINER
WOVEN - 100% COTTON

STROBEL
NON-WOVEN - POLYESTER

MIDSOLE
TPU

OUTSOLE
TPU

MANUFACTURING

CUT + SEW
UPPER CONSTRUCTION

3D PRINTING
MIDSOLE & OUTSOLE

ASSEMBLY
ADHESIVE GLUE

FEEDBACK

LIKES

- Stylish
- Chelsea style for easy slip-on
- Lightweight for summer & hot weather
- Natural materials - sustainable

CONCERNS

- Should boot be taller for increased protection
- Being able to use the strap with beekeeping gloves on
- If DWR coating will truly repel moisture in outdoor conditions
- If there is enough padding/thickness around the foot



*Vizcom Rendering

SUSTAINABILITY METRIC

PF CLOVER
SUSTAINABLE UTILITY BOOT



CO_2e / pair
~ 6.7 kg

Primary Upper Material
Organic Cotton

THE GROUNDER
HERITAGE FOOTWEAR



CO_2e / pair
~ 8.4 kg

Primary Upper Material
Cotton Canvas

BLUNDSTONE CHELSEA
LIFESTYLE BOOT



CO_2e / pair
~ 15 kg

Primary Upper Material
Leather

REFLECTION

WHAT I LEARNED

- Cotton is best used in footwear as a non-structural material due to limited shape retention
- Cotton requires intentional patterning to prevent collapse
- DWR in cotton improved water resistance but came with broader sustainability tradeoffs
- Scanning the upper improved fit accuracy when modeling the midsole & outsole
- Cotton is most effective in comfort-focused footwear applications rather than in high-performance




SAVE
THE BEES
... Keeper's Ankles